

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	K. Ragavi	Reg. No.:	192524364
Section / Batch:	B. Tech AI&DS/2 ND year	Date:	17/07/2026

Code of Conduct

I K. Ragavi (192524364) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Ragavi

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

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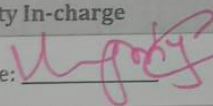
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Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. convert the expression $((a+b) + c * (d+e) + f) * (g+h)$ to 1) Prefix expression, 2) postfix expression.

Step 1: Identify the structure

The expression is: $((a+b) + (c * (d+e))) + f) * (g+h)$

Step 2: convert inner sub-expressions

$(a+b) \rightarrow$ Prefix: $+ab$ Postfix: $ab+$

$(d+e) \rightarrow$ Prefix: $+de$ Postfix: $de+$

$c * (d+e) \rightarrow$ Prefix: $*c+de$ Postfix: $de+c*$

$(a+b) + c * (d+e) \rightarrow$ Prefix: $++ab*c+de$ Postfix: $ab+de+c*+$

$\dots + f \rightarrow$ Prefix: $+++ab*c+def$ Postfix: $ab+de+c*+f+$ $(g+h)$

$\rightarrow$ Prefix: $+gh$ Postfix: $gh+$

Step 3: combine with outer multiplication

Prefix Expression: $*+++ab*c+def+gh$

Postfix Expression: $ab+de+c*+f+gh+*$

2. Perform the binary search in an array of n elements using recursion function, in which a function has to call by itself use these elements given in sorted order and assume 88 as the key element. [23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 88]

Recursive binary search procedure

Formula : $\text{mid} = (\text{low} + \text{high}) / 2$ (integer division)

If $\text{arr}[\text{mid}] = \text{key} \rightarrow$ element found.

If $\text{arr}[\text{mid}] < \text{key} \rightarrow$ search right half : call Binary search(arr , $\text{mid} + 1$, high , key).

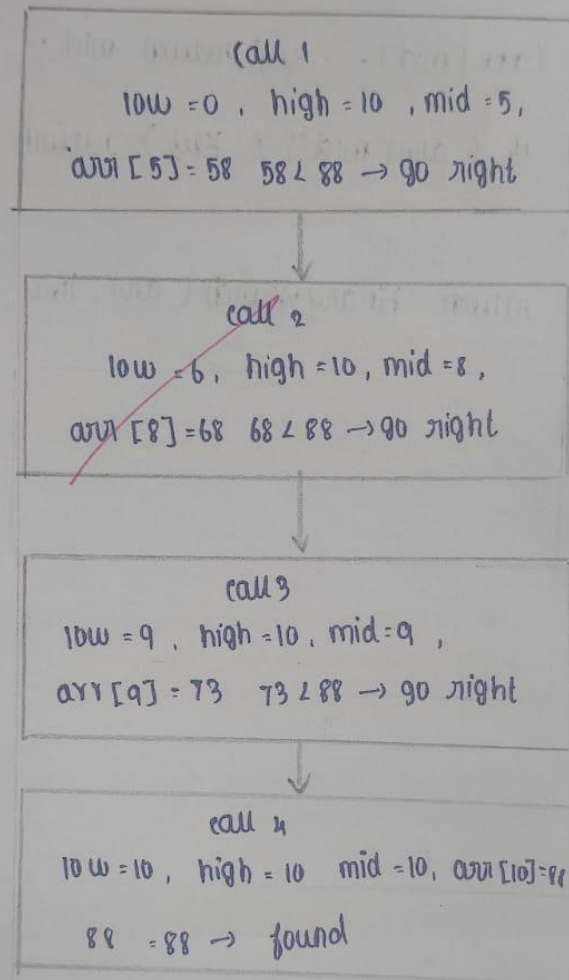
If $\text{arr}[\text{mid}] > \text{key} \rightarrow$ search left half : call Binary search(arr , low , $\text{mid} - 1$, key).

Trace of recursive call

call	low	high	mid	arr[mid]	Action
1	0	10	5	58	$58 < 88$, go right ($\text{low} = 6$)
2	6	10	8	68	$68 < 88$, go right ($\text{low} = 9$)
3	9	10	9	73	$73 < 88$, go right ($\text{low} = 10$)
4	10	10	10	88	$88 = 88$, found at index 10

Recursive call tree diagram

Recursive call tree - binary search for key = 88



Result

The key element 88 is found at index 10 after 4 recursive calls.

Recursive function (reference)

```
int binarysearch (int arr[], int low, int high, int key) {  
    if (low > high) return -1;  
    int mid = (low + high) / 2;  
    if (arr[mid] == key) return mid;  
    else if (arr[mid] < key) return binarysearch(arr, mid + 1,  
high, key);  
    else return binarysearch(arr, low, mid - 1, key);  
}
```

Uppin

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